

POWER SYSTEM 2 – DETAILED THEORY NOTES

MODULE 1: POWER FLOW ANALYSIS

1.1 Introduction to Power System Structure

Components: Generators, Transformers, Transmission Lines, Loads, Shunt/Series Compensators

Bus Classification:

Slack Bus (Swing Bus): A slack bus (also called swing bus) is a special bus in a power system used in load-flow analysis where the voltage magnitude and phase angle are specified, and it supplies or absorbs the remaining real and reactive power required to balance the system losses.

Reference bus, voltage magnitude and angle specified, real and reactive power unknown

PV Bus (Generator Bus): A PV bus, also called a generator bus, is a bus in a power system where the real power (P) and voltage magnitude (|V|) are specified, while the reactive power (Q) and phase angle (δ) are calculated during load-flow analysis.

Real power and voltage magnitude specified, reactive power and voltage angle unknown

PQ Bus (Load Bus): A PQ bus, also called a load bus, is a bus in a power system where the real power (P) and reactive power (Q) are specified, while the voltage magnitude (|V|) and phase angle (δ) are calculated during load-flow analysis.

Real and reactive power specified, voltage magnitude and angle unknown

1.2 Bus Admittance Matrix (Y_{bus}) Formation

Diagonal Elements (Y_{ii}): Sum of admittances connected to bus i

Off-diagonal Elements (Y_{ij}): Negative of admittance between bus i and j

Properties:

Symmetric if no phase shifters

Sparse for large systems

Complex matrix: $Y_{ij} = G_{ij} + jB_{ij}$

1.3 Power Flow Equations

For bus i:

$$P_i = V_i \sum_{j=1}^n V_j [G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)]$$

$$Q_i = V_i \sum_{j=1}^n V_j [G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j)]$$

1.4 Numerical Solution Methods

Gauss-Seidel Method

Algorithm:

Assume flat start ($V=1\angle 0^\circ$ for all buses except slack)

Update voltage iteratively:

$$V_i^{(k+1)} = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{V_i^{(k)*}} - \sum_{j=1}^{i-1} Y_{ij} V_j^{(k+1)} - \sum_{j=i+1}^n Y_{ij} V_j^{(k)} \right]$$

Advantages: Simple, less memory

Disadvantages: Slow convergence, poor for ill-conditioned systems

Newton-Raphson Method

Formulation: Solve nonlinear equations using Taylor series expansion

Mismatch Equations:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J_1 & J_2 \\ J_3 & J_4 \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta |V| \end{bmatrix}$$

Where:

- $J_1 = \partial P / \partial \theta$, $J_2 = \partial P / \partial |V|$
- $J_3 = \partial Q / \partial \theta$, $J_4 = \partial Q / \partial |V|$

Advantages: Quadratic convergence, reliable

Disadvantages: Requires Jacobian computation and inversion

1.5 Computational Issues in Large Systems

Sparse Matrix Techniques: Store only non-zero elements

Fast Decoupled Load Flow: Assumes

$$\cos \theta_{ij} \approx 1, \sin \theta_{ij} \approx 0, G_{ij} \sin \theta_{ij} \ll B_{ij}$$

Separates P- θ and Q-V equations

Uses constant Jacobian matrices

NUMERICALS:

1. Ybus Formation with Shunt Elements

Q: For a 3-bus system:

Line 1-2: $z = 0.02 + j0.08$ pu, $y_{sh}/2 = j0.01$ pu

Line 1-3: $z = 0.01 + j0.04$ pu, $y_{sh}/2 = j0.005$ pu

Line 2-3: $z = 0.03 + j0.12$ pu, $y_{sh}/2 = j0.015$ pu

Form Ybus.

A:

$$y_{12} = 1/(0.02 + j0.08) = 2.94 - j11.76$$

$$y_{13} = 1/(0.01 + j0.04) = 5.88 - j23.53$$

$$y_{23} = 1/(0.03 + j0.12) = 1.96 - j7.84$$

$$Y_{11} = y_{12} + y_{13} + j0.01 + j0.005 = 8.82 - j35.275$$

$$Y_{22} = y_{12} + y_{23} + j0.01 + j0.015 = 4.90 - j19.575$$

$$Y_{33} = y_{13} + y_{23} + j0.005 + j0.015 = 7.84 - j31.345$$

$$Y_{12} = Y_{21} = -2.94 + j11.76$$

$$Y_{13} = Y_{31} = -5.88 + j23.53$$

$$Y_{23} = Y_{32} = -1.96 + j7.84$$

2. Gauss-Seidel - 2 Bus System

Q: Bus1: Slack, $V_1 = 1.05 \angle 0^\circ$

Bus2: PQ, $P_2 = -0.6$, $Q_2 = -0.3$ pu

$y_{12} = 4 - j16$ pu

Perform 2 iterations.

A:

$$\text{Iter1: } V_2 = 0.962 - 0.022j = 0.962 \angle -1.31^\circ$$

$$\text{Iter2: } V_2 = 0.960 - 0.025j = 0.960 \angle -1.49^\circ$$

3. Gauss-Seidel - 3 Bus System

Q: 3-bus system with:

Bus1: Slack, $1.05 \angle 0^\circ$

Bus2: PQ, $-0.8-j0.4$
Bus3: PV, $P=0.6$, $|V|=1.04$
Ybus as problem 1
1 iteration.

A:

$$V_2 = 0.978 - 0.035j = 0.978 \angle -2.05^\circ$$

$$Q_3 = 0.312 \text{ pu}$$

$|V_3|$ maintained at 1.04

4. Newton-Raphson - First Iteration

Q: 2-bus system from problem 2. Perform 1 NR iteration.

A:

$$\Delta\theta_2 = -0.026 \text{ rad}, \Delta V_2 = -0.038 \text{ pu}$$

$$V_2 = 0.962 \angle -1.49^\circ$$

5. Newton-Raphson - Jacobian Formation

Q: For 2-bus system, calculate all Jacobian elements at flat start.

A:

$$J_{11} = \partial P_2 / \partial \theta_2 = -16$$

$$J_{12} = \partial P_2 / \partial V_2 = 4$$

$$J_{21} = \partial Q_2 / \partial \theta_2 = 4$$

$$J_{22} = \partial Q_2 / \partial V_2 = -16$$

6. Line Flow Calculation

Q: Line with $z=0.01+j0.04 \text{ pu}$
 $V_s=1.02 \angle 5^\circ$, $V_r=0.98 \angle -2^\circ$
Find P, Q at both ends.

A:

$$P_{sr} = 0.423 \text{ pu}, Q_{sr} = 0.156 \text{ pu}$$

$$P_{rs} = -0.418 \text{ pu}, Q_{rs} = 0.142 \text{ pu}$$

Loss = $0.005 + j0.014$ pu

7. Slack Bus Power

Q: From solved system:

$V_1 = 1.05 \angle 0^\circ$, $V_2 = 0.98 \angle -4^\circ$, $V_3 = 1.01 \angle -2^\circ$

Line flows known. Find P_1 , Q_1 .

A:

$P_1 = 0.823$ pu, $Q_1 = 0.342$ pu

8. Voltage Correction

Q: PV bus: $|V| = 1.04$ specified

After iteration: $V = 1.06 \angle -3^\circ$

What V used in next iteration?

A: $V = 1.04 \angle -3^\circ$

9. Reactive Power Limit

Q: Generator Q limits: -0.2 to 0.5 pu

Solution gives $Q = 0.55$ pu

How proceed?

A: Treat as PQ bus with $Q = 0.5$ pu fixed.

10. Convergence Check

Q: NR iteration: $\max|\Delta P| = 0.0008$, $\max|\Delta Q| = 0.0012$

Tolerance = 0.001

Continue?

A: Yes, $\Delta Q > \text{tolerance}$.

11. Fast Decoupled Load Flow

Q: $B' = -50$, $\Delta P = -0.2$

Find $\Delta\theta$.

A: $\Delta\theta = -0.2/(-50) = 0.004 \text{ rad}$

12. Transformer Tap

Q: Transformer 132/33 kV, $\pm 10\%$ taps
Need 31.5 kV secondary
Which tap?

A: Tap at +5% (ratio=4.19 vs 4.0 normal)

13. Shunt Capacitor

Q: Load: 50 MW at 0.8 lag
Improve pf to 0.95 lag
Capacitor size?

A: $Q_c = 50(\tan 36.87^\circ - \tan 18.19^\circ) = 21.0 \text{ MVAR}$

14. Loss Calculation

Q: $P_{\text{gen}} = 1.2 + 0.9 \text{ pu}$, $P_{\text{load}} = 0.8 + 0.7 + 0.6 \text{ pu}$
Find losses.

A: $P_{\text{loss}} = 2.1 - 2.1 = 0$ (should be small positive)

15. Multi-V bus

Q: 4-bus system: 2 PV, 1 PQ, 1 Slack
How many unknowns in NR?

A: 2 voltage angles + 1 voltage magnitude = 3 unknowns

MODULE 2: POWER SYSTEM STABILITY ANALYSIS

2.1 Basic Concepts

Stability: Ability of power system to maintain synchronism under normal and disturbed conditions

Types:

Rotor Angle Stability: Synchronous generators remain in synchronism

Voltage Stability: Maintain acceptable voltage at all buses

Frequency Stability: Maintain frequency within limits

2.2 Swing Equation Derivation

From Newton's second law for rotating masses:

$$J \frac{d^2 \theta_m}{dt^2} = T_m - T_e$$

Converting to electrical angles and per-unit:

$$\frac{2H}{\omega_s} \frac{d^2 \delta}{dt^2} = P_m - P_e$$

Where:

H = Inertia constant (MW-s/MVA)

δ = Rotor angle (electrical radians)

P_m = Mechanical power input

P_e = Electrical power output

2.3 Power-Angle Relationship

For simple system (generator→transformer→line→infinite bus):

$$P_e = \frac{E'V}{X_{eq}} \sin \delta$$

Where ,

$$X_{eq} = X_d' + X_t + X_{line}$$

2.4 Equal Area Criterion

Applied to: Single machine infinite bus system

Principle: Stability if accelerating area = decelerating area

Critical Clearing Angle (δ_{cr}):

$$\int_{\delta_0}^{\delta_{cr}} (P_m - P_{e,fault})d\delta = \int_{\delta_{cr}}^{\delta_{max}} (P_{e,post-fault} - P_m)d\delta$$

Critical Clearing Time: Corresponding time for δ_{cr}

2.5 Numerical Integration Methods

Modified Euler Method

$$\delta_{n+1} = \delta_n + \Delta t \cdot \omega_n + \frac{\Delta t^2}{2} \cdot \frac{d\omega}{dt} \Big|_n$$

$$\omega_{n+1} = \omega_n + \frac{\Delta t}{2} \left(\frac{d\omega}{dt} \Big|_n + \frac{d\omega}{dt} \Big|_{n+1} \right)$$

Runge-Kutta 4th Order

More accurate but computationally intensive

2.6 Stability Enhancement Techniques

Fast Fault Clearing: Reduce fault duration

High Speed Reclosure: Quickly restore lines

Excitation Control: Increase transient stability limit

Series Compensation: Reduce effective line reactance

Braking Resistors: Absorb excess energy during faults

Generation Rescheduling: Shift generation to critical lines

NUMERICALS:

1. Inertia Constant

Q: 150 MVA generator, $WR^2=2 \times 10^6 \text{ lb}\cdot\text{ft}^2$, speed=3600 rpm
Find H.

A:

$$KE = 0.5 \times (2 \times 10^6 / 32.2) \times (2\pi \times 3600 / 60)^2 \times (1/550) \text{ ft}\cdot\text{lb/s}$$

$$= 148.6 \text{ MJ}$$

$$H = 148.6 / 150 = 0.991 \text{ MJ/MVA}$$

2. Swing Equation Formulation

Q: $H=4 \text{ MJ/MVA}$, $f=50 \text{ Hz}$
Find M.

A: $M = 2 \times 4 / (100\pi) = 0.0255 \text{ pu}$

3. Initial Power Angle

Q: $E'=1.3$, $V=1.0$, $X=0.9 \text{ pu}$, $P=1.0 \text{ pu}$
Find δ_0 .

A: $\delta_0 = \sin^{-1}(1.0 \times 0.9 / 1.3) = 43.6^\circ$

4. Equal Area - Simple

Q: $P_m=1.0$, $P_{\max}=1.5$, $\delta_0=30^\circ$, fault reduces P_e to 0
Find $\delta_{\max}$.

A: $\delta_{\max} = 180^\circ - 41.8^\circ = 138.2^\circ$

5. Equal Area - Critical Angle

Q: From above, find δ_{cr} .

A: Solve: $1.0(\delta_{cr}-0.5236)=1.5(\cos\delta_{cr}-\cos138.2^\circ)-1.0(2.412-\delta_{cr})$
 $\delta_{cr}=1.18 \text{ rad}=67.6^\circ$

6. Euler Method

Q: $\delta_0=0.5 \text{ rad}$, $\omega_0=0$, $P_a=0.6 \text{ pu}$, $M=0.03$
 $\Delta t=0.1 \text{ s}$, find δ_1 .

A: $\alpha=0.6/0.03=20 \text{ rad/s}^2$
 $\delta_1=0.5+0+0.5\times 20\times 0.01=0.6 \text{ rad}$

7. Modified Euler

Q: Same data, find δ_1 using Modified Euler.

A:

Predictor: $\omega_p=2.0$, $\delta_p=0.7$

$\alpha_p=20$

Corrector: $\omega_1=2.0$, $\delta_1=0.7$

8. Runge-Kutta 2nd Order

Q: Same data, RK2.

A:

$k1_\omega=20\times 0.1=2$, $k1_\delta=0\times 0.1=0$

$\omega_{temp}=0+2/2=1$, $\delta_{temp}=0.5+0/2=0.5$

$k2_\omega=20\times 0.1=2$, $k2_\delta=1\times 0.1=0.1$

$\omega_1=0+2=2$, $\delta_1=0.5+0.1=0.6$

9. Critical Clearing Time

Q: $\delta_{cr}=80^\circ=1.396 \text{ rad}$, $\delta_0=30^\circ=0.524 \text{ rad}$
 $P_a=0.8 \text{ pu}$, $M=0.025$
Find t_{cr} .

A:

$$\delta_{cr}-\delta_0=0.5(P_a/M)t_{cr}^2$$

$$0.872=0.5(0.8/0.025)t_{cr}^2$$

$$t_{cr}=\sqrt{(0.872 \times 0.025/0.4)}=0.234 \text{ s}$$

10. Multi-Machine COA

Q: 3 machines: $H_1=3$, $H_2=4$, $H_3=5$
 $\delta_1=25^\circ$, $\delta_2=30^\circ$, $\delta_3=35^\circ$
Find δ_{COA} .

A: $\delta_{COA}=(3 \times 25 + 4 \times 30 + 5 \times 35)/12=30.8^\circ$

11. Transient Stability Limit

Q: With clearing time=0.2s, what % of P_{max} can be transmitted?
 $H=5$, $X=0.8$

A: Approximately 70% of P_{max}

12. Effect of Series Compensation

Q: Series compensation reduces X from 0.8 to 0.6 pu
How does P_{max} change?

A: P_{max} increases by $(0.8/0.6-1) \times 100\%=33.3\%$

13. Stability with AVR

Q: With AVR, E' increases from 1.2 to 1.3 pu
Original $P_{max}=1.5 \text{ pu}$
New P_{max} ?

A: New $P_{max}=1.5 \times (1.3/1.2)=1.625 \text{ pu}$

14. Machine Separation

Q: Area1 gen=1000 MW, Area2 load=800 MW
Tie-line trips, $H_{eq}=4$
Initial df/dt ?

A: $df/dt = -(800/1000)/(2 \times 4) = -0.1 \text{ pu/s} = -5 \text{ Hz/s}$

15. Damping Coefficient

Q: Oscillation decays to 50% in 5 cycles
Find damping ratio.

A: $\zeta \approx 0.11$

MODULE 3: FREQUENCY AND VOLTAGE CONTROL

3.1 Frequency Control Hierarchy

Primary Control (Droop Control)

Governor Action: Local, fast (seconds)

Droop Characteristic: $R = -(\Delta f / \Delta P) (\%)$

Power Sharing: Generators share load change inversely proportional to droop

Secondary Control (AGC)

Area Control Error (ACE): $ACE = \Delta P_{tie} + B \Delta f$

Objective: Zero ACE, restore frequency

Tertiary Control

Economic dispatch, unit commitment

3.2 Automatic Generation Control (AGC)

Components: Tie-line bias control, supplementary control

Control Loop:

Measure ACE

Send control signals to generators

Adjust governor setpoints

3.3 Reactive Power Control

Excitation Systems

Purpose: Control terminal voltage, reactive power output

Components: Exciter, voltage regulator, stabilizers

AVR Block Diagram: Reference voltage → Comparator → Amplifier → Exciter → Generator

Static VAR Compensators (SVC)

Types: TCR, TSC, FC-TCR

Characteristic: Slope similar to synchronous condenser

Control: Maintain voltage by absorbing/supplying reactive power

STATCOM

Advantages over SVC: Faster response, better low-voltage performance

Operation: Voltage source converter generates/absorbs reactive power

3.4 Transformer Tap Changing

Objective: Maintain secondary voltage within limits

Types: On-load tap changers (OLTC), off-circuit tap changers

Control: Automatic or manual, line drop compensation

3.5 Power Flow Control Devices

Phase Shifting Transformers: Control real power flow

Unified Power Flow Controller (UPFC): Combined series and shunt compensation

HVDC Links: Controllable power transfer, asynchronous interconnections

NUMERICALS:

1. Governor Droop

Q: 200 MW gen, 4% droop
Frequency drop from no-load to full-load?

A: $\Delta f = 0.04 \times 50 = 2$ Hz

2. Load Sharing

Q: Two gens: 200 MW (3% droop) and 300 MW (4% droop)
Load=400 MW at 50 Hz
Find sharing.

A:

$$R1 = 0.03 \times 50 / 200 = 0.0075 \text{ Hz/MW}$$

$$R2 = 0.04 \times 50 / 300 = 0.00667 \text{ Hz/MW}$$

$$P1 = 400 \times 0.00667 / (0.0075 + 0.00667) = 188 \text{ MW}$$

$$P2 = 212 \text{ MW}$$

3. System Frequency

Q: From above, no-load freq=51 Hz
Find system frequency.

A: $f = 51 - 0.0075 \times 188 = 49.59$ Hz

4. AGC - ACE

Q: Area: $B = 0.1$ pu/Hz, $\Delta f = -0.05$ Hz
 $\Delta P_{\text{tie}} = +0.02$ pu (export)
Base=1000 MW
Find ACE.

A: $ACE=0.02+0.1\times(-0.05)=0.015$ pu=15 MW

5. AGC Response

Q: ACE=20 MW, $K_I=0.15$, $\Delta t=2$ s
Generation change?

A: $\Delta P=0.15\times 20\times 2=6$ MW

6. Load Frequency Control

Q: System: $R=0.05$ pu, $D=0.8$ pu/Hz
Load increase=0.1 pu
Steady-state Δf ?

A: $\Delta f=-0.1/(1/R+D)=-0.1/(20+0.8)=-0.0048$ pu=-0.24 Hz

7. SVC Rating

Q: Load varies between 50+ j40 to 50-j30 MVA
Find SVC rating.

A: Rating=40-(-30)=70 MVAR

8. STATCOM Current

Q: STATCOM at 400 kV, ± 100 MVAR
Current rating?

A: $I=100/(\sqrt{3}\times 400)=0.1443$ kA=144.3 A

9. Shunt Capacitor

Q: Load: 20 MW at 0.75 lag
Improve to 0.95 lag
Capacitor size?

A: $Q_c = 20(\tan 41.4^\circ - \tan 18.2^\circ) = 20 \times 0.858 = 17.16 \text{ MVAR}$

10. Voltage Drop

Q: Line: $R=0.1$, $X=0.4$ pu
 $P=0.8$, $Q=0.6$ pu at receiving end
Voltage drop?

A: $\Delta V \approx (0.8 \times 0.1 + 0.6 \times 0.4) / 1 = 0.08 + 0.24 = 0.32 \text{ pu}$

11. Tap Changer

Q: Transformer 132/11 kV
Need 10.8 kV secondary
Which tap? ($\pm 10\%$, 16 steps)

A: Need ratio $= 132 / 10.8 = 12.22$ (normal $= 12$)
% change $= (12.22 - 12) / 12 = 1.83\%$
Taps $= 1.83 / 0.625 = 2.93 \approx 3$ taps raise

12. Phase Shifter

Q: Phase shifter adds 5° shift
Line $X=0.2$ pu
Power flow change?

A: $\Delta P \approx (1/0.2) \times (5 \times \pi / 180) = 5 \times 0.0873 = 0.4365 \text{ pu}$

13. Reactive Power from Generator

Q: Generator: $X_d=1.5$ pu, $P=0.9$ pu, $|V_t|=1.05$, $|E_f|=1.8$
Find Q .

A:

$$P = (1.8 \times 1.05 / 1.5) \sin \delta = 1.26 \sin \delta = 0.9$$

$$\sin \delta = 0.7143, \delta = 45.6^\circ$$

$$Q = (1.8 \times 1.05 \cos 45.6^\circ - 1.05^2) / 1.5 = 0.312 \text{ pu}$$

14. AVR Setting

Q: Generator terminal voltage drops from 1.05 to 1.0 pu
AVR increases field to restore voltage
If excitation increases by 20%, new terminal voltage?

A: Approximately $V_t \propto \sqrt{E_f}$
New $V_t \approx 1.05 \times \sqrt{1.2} = 1.149$ pu

15. Power Factor Correction Cost

Q: Load: 1000 kW at 0.7 lag for 8000 hrs/year
Capacitor cost: ₹500/kVAR
Energy cost: ₹5/kWh
Payback period?

A:

$$Q_c = 1000(\tan 45.6^\circ - \tan 18.2^\circ) = 1000 \times 0.858 = 858 \text{ kVAR}$$

$$\text{Cap cost} = 858 \times 500 = ₹4,29,000$$

$$\text{Loss reduction} = 1000 \times (1/0.7 - 1/0.95) = 376 \text{ kW}$$

$$\text{Annual savings} = 376 \times 8000 \times 5/1000 = ₹15,04,000$$

$$\text{Payback} = 4,29,000/15,04,000 = 0.285 \text{ years} \approx 3.4 \text{ months}$$

MODULE 4: GENERATION ECONOMICS AND CONTROL

4.1 Generator Cost Characteristics

Input-Output Curve: Fuel input vs. power output

Heat Rate Curve: BTU/kWh vs. MW

Incremental Fuel Cost Curve: dF/dP (\$/MWh) vs. MW

4.2 Economic Dispatch Problem

Objective: Minimize

$$C_T = \sum_{i=1}^N C_i(P_{gi})$$

Subject to:

Power balance: $\sum P_{gi} = PD + PL$

Generation limits:

$$P_{gi}^{min} \leq P_{gi} \leq P_{gi}^{max}$$

4.3 Solution Methods

Lambda Iteration Method

Assume initial λ

Calculate

$$P_{gi} \text{ from: } \frac{dC_i}{dP_{gi}} = \lambda$$

Check power balance

Adjust λ using gradient method

Repeat until convergence

Penalty Factors (Loss Formula)

Modified Lagrangian:

$$L = \sum C_i(P_{gi}) + \lambda(P_D + P_L - \sum P_{gi})$$

Optimality Condition:

$$\frac{dC_i}{dP_{gi}} \cdot L_i = \lambda, \quad \text{where } L_i = \frac{1}{1 - \frac{\partial P_L}{\partial P_{gi}}}$$

4.4 Electricity Markets

Market Models

Vertically Integrated: Single entity owns generation, transmission, distribution

Purchasing Agency: Single buyer model

Wholesale Competition: Generators compete, retailers serve customers

Retail Competition: End-users choose suppliers

Spot Pricing

Locational Marginal Price (LMP): Cost to serve next MW at specific location

Components: Energy, congestion, loss components

4.5 Ancillary Services

Frequency Regulation: AGC service

Spinning Reserve: Synchronized, available within 10 minutes

Non-spinning Reserve: Can be synchronized within 10 minutes

Reactive Power Support: Voltage control

Black Start: Restart system after blackout

4.6 Demand Side Management (DSM)

Load Shifting: Move load from peak to off-peak

Peak Clipping: Reduce peak demand

Valley Filling: Increase off-peak consumption

Strategic Conservation: Reduce overall consumption

4.7 Control Center Functions

SCADA System

Components: RTUs, MTU, communication links, HMI

Functions: Data acquisition, monitoring, control, alarm processing

Phasor Measurement Units (PMUs)

GPS-synchronized: Time-stamped measurements

Measurements: Voltage/current phasors, frequency, rate of change of frequency

Advantages: High resolution, time-synchronized, wide-area visibility

Wide-Area Measurement Systems (WAMS)

Architecture: PMUs → PDCs → Control center

Applications: Oscillation detection, stability monitoring, fault location

4.8 State Estimation

Objective: Estimate system state (V , θ) from measurements

Mathematical Formulation:

Solution: Weighted least squares using Newton's method

$$z = h(x) + e$$

$$\min J(x) = [z - h(x)]^T R^{-1} [z - h(x)]$$

4.9 Security Assessment

Contingency Analysis: "What-if" analysis for N-1, N-2 outages

Preventive Control: Take action before contingency to maintain security

Emergency Control: Take action after contingency to restore system

4.10 AI Applications in Power Systems

Load Forecasting: Neural networks for short-term forecasting

Fault Diagnosis: Expert systems for fault classification

Stability Prediction: Machine learning for transient stability assessment

PMU Data Analytics: Pattern recognition for oscillation detection

Market Bidding Strategies: Reinforcement learning for optimal bidding

NUMERICALS:

1. Cost Curve Parameters

Q: $C(P) = aP^2 + bP + c$

Given: At $P = 100$ MW, $C = ₹8000/\text{hr}$, $dC/dP = ₹60/\text{MWh}$

At $P = 150$ MW, $dC/dP = ₹70/\text{MWh}$

Find a, b, c .

A:

$$dC/dP = 2aP + b$$

$$\text{At } 100 \text{ MW: } 60 = 200a + b$$

$$\text{At } 150 \text{ MW: } 70 = 300a + b$$

$$\text{Solve: } a = 0.1, b = 40$$

$$\text{At } P = 100: 8000 = 0.1 \times 10000 + 40 \times 100 + c$$

$$c = 8000 - 1000 - 4000 = 3000$$

2. Economic Dispatch - 2 Units

Q: $C_1 = 0.01P_1^2 + 20P_1 + 100$

$C_2 = 0.015P_2^2 + 18P_2 + 120$

Load = 200 MW

Find optimal dispatch.

A:

$$0.02P_1 + 20 = 0.03P_2 + 18 = \lambda$$

$$P_1 + P_2 = 200$$

$$\text{Solve: } P_1 = 120 \text{ MW, } P_2 = 80 \text{ MW, } \lambda = ₹22.4/\text{MWh}$$

3. Economic Dispatch - 3 Units

Q: Add $C_3 = 0.02P_3^2 + 15P_3 + 150$

Load = 300 MW

Find dispatch.

A:

$$0.02P_1 + 20 = 0.03P_2 + 18 = 0.04P_3 + 15 = \lambda$$

$$P_1 + P_2 + P_3 = 300$$

$$\text{Solve: } P_1 = 138.5, P_2 = 107.7, P_3 = 53.8, \lambda = ₹22.77/\text{MWh}$$

4. Lambda Iteration with Limits

Q: From above, limits: $P_1 \leq 100$, $P_2 \leq 150$, $P_3 \leq 100$

Find dispatch.

A:

P_1 clamps at 100 MW

$$\text{Then: } 0.03P_2 + 18 = 0.04P_3 + 15$$

$$P_2 + P_3 = 200$$

$$\text{Solve: } P_2 = 114.3, P_3 = 85.7$$

$$\lambda = ₹21.43/\text{MWh}$$

5. Penalty Factors

Q: Loss formula: $P_L = 0.001P_1^2 + 0.002P_2^2$

At $P_1 = 100$, $P_2 = 150$

Find penalty factors.

A:

$$\partial P_L / \partial P_1 = 0.002 \times 100 = 0.2$$

$$L_1 = 1 / (1 - 0.2) = 1.25$$

$$\partial P_L / \partial P_2 = 0.004 \times 150 = 0.6$$

$$L_2 = 1 / (1 - 0.6) = 2.5$$

6. Economic Dispatch with Losses

Q: Using above penalty factors, IC1=22, IC2=21
Find λ .

A:

$$IC1 \times L1 = IC2 \times L2 = \lambda$$

$$22 \times 1.25 = 27.5 = \lambda$$

$$21 \times 2.5 = 52.5 \text{ (not equal)}$$

Need iterative adjustment of P1, P2

7. B-Coefficients

Q: Loss formula: $P_L = B_{11}P_1^2 + B_{22}P_2^2 + 2B_{12}P_1P_2$

Given: $B_{11}=0.001$, $B_{22}=0.002$, $B_{12}=0.0005$

At $P_1=100$, $P_2=150$

Find losses.

A:

$$P_L = 0.001 \times 10000 + 0.002 \times 22500 + 2 \times 0.0005 \times 15000$$

$$= 10 + 45 + 15 = 70 \text{ MW}$$

8. Spot Price Calculation

Q: Marginal cost = 25 ₹/MWh

Congestion = 3 ₹/MWh

Loss = 2 ₹/MWh

Find LMP.

A: $LMP = 25 + 3 + 2 = 30 \text{ ₹/MWh}$

9. Ancillary Services Cost

Q: Spinning reserve: 200 MW at ₹10/MW-hr

Regulation: 50 MW at ₹15/MW-hr

Daily cost?

A: Daily cost = $200 \times 10 \times 24 + 50 \times 15 \times 24$

$$= 48,000 + 18,000 = ₹66,000$$

10. SCADA Data Rate

Q: 50 RTUs, 100 points each, scan every 2s
32 bits/point
Required bandwidth?

A: Data rate = $50 \times 100 \times 32 / 2 = 80,000$ bps

11. PMU Placement

Q: 8-bus system, each PMU sees bus and adjacent buses
Minimum PMUs for full observability?

A: Typically 3 PMUs for 8-bus system

12. State Estimation

Q: Measurements: $V = 1.02 \pm 0.01$, $P = 0.85 \pm 0.02$
Weighted estimate?

A: Weights: $w_1 = 1/0.01^2 = 10000$, $w_2 = 1/0.02^2 = 2500$
Weighted average if both measure same quantity

13. Bad Data Detection

Q: Measurements: $z_1 = 1.02$, $z_2 = 1.03$, $z_3 = 1.01$
All accuracy ± 0.01
Is any bad data?

A: All within $3\sigma = 0.03$, so acceptable

14. Contingency Ranking

Q: Line outages cause overloads:
Outage1: 120% loading
Outage2: 115% loading
Outage3: 105% loading
Rank severity.

A: Outage1 > Outage2 > Outage3

15. Reliability Indices

Q: System: 1000 customers
10 outages, total 50 hrs, 5000 customer-hours interrupted
Find SAIFI, SAIDI.

A:

SAIFI = $10/1000 = 0.01$ interruptions/customer

SAIDI = $5000/1000 = 5$ hrs/customer

IMPORTANT DERIVATIONS FOR EXAMS

Derivation of Swing Equation from Newton's second law

Equal Area Criterion derivation for critical clearing angle

Ybus formation from network data

Newton-Raphson Jacobian matrix elements derivation

Lambda iteration with transmission losses

Area Control Error (ACE) equation derivation

Droop characteristic and power sharing derivation

State estimation weighted least squares formulation